DEEP PATEL

pateldeep1842@gmail.com | linkedin.com/in/deeppatel1842 | github.com/deeppatel1842
Seattle, WA | +1 (425) 305-1382

SUMMARY

AI Engineer with an M.S. in Computer Science and hands-on experience building LLM, RAG, AI agent, and transformer fine-tuning projects. Skilled in Python, FastAPI, vector search, reranking, citation-aware generation, and evaluation workflows for grounded AI systems.

EDUCATION

University of Oklahom — Aug 2023 - May 2025
MS in Computer Science
Coursework: Natural Language Processing, Artificial Intelligence, Machine Learning, Transformer Models

Indian Institute of Information Technology (IIIT), India — Aug 2018 – May 2022
BTech in Computer Science and Engineering

RESEARCH PROJECTS

TRACE: Claim Verification & Cost Control Layer for LLM Systems [GitHub] May 2026 – Present

- Built a modular trust layer for LLM and RAG systems with guarded input routing, risk classification, evidence retrieval, claim verification, and final answer repair.
- Implemented CAGE, a claim-verification workflow that extracts atomic claims, retrieves supporting evidence, checks citation support, and removes unsupported statements before final output.
- Designed CostGuard to control model routing, retrieval depth, verification depth, compression level, token usage, and per-query cost before generation and verification.
- Added domain-scoped memory with injection screening, verified-claim storage, TTL-based re-verification, and cache invalidation to keep stored knowledge safer and fresher.

EXPERIENCE

Machine Learning Engineer Volunteer — Aug 2025 – Present
Community Dreams Foundation, Remote, USA

- Supported machine learning application development by analyzing datasets, preparing data workflows, and identifying model improvement opportunities for community-focused projects.

AI/ML Engineer — May 2022 – May 2023
Firenix Technologies, Pune, India

- Collaborated with senior engineers to build computer vision pipelines for image classification, face recognition, and anomaly-detection workflows using Python and PyTorch.
- Improved data preprocessing and inference workflows by standardizing image cleaning, resizing, batching, and prediction handling for more reliable model outputs.
- Supported model testing and backend integration by packaging prediction workflows into reusable scripts and API-ready components.

PROJECTS

Agentic Research Assistant [GitHub] Jan 2026 – May 2026

- Architected a planner-based research system that routes user queries across document Q&A and web research workflows using FastAPI, tool execution, and shared trace logging.
- Developed a document RAG and evaluation pipeline with PDF ingestion, embeddings, reranking, citation-based generation, Tavily/arXiv/Wikipedia tools, token-cost tracking, retrieval metrics, and LLM-as-judge scoring.

Privacy-Aware Financial Agent Backend [GitHub] Sep 2025 – Dec 2025

- Designed a deterministic-first financial AI pipeline that routes questions through intent detection, risk tiering, policy planning, compute, validation, grounding, and LLM synthesis.
- Scaffolded a FastAPI and Pydantic backend with modular sections for secure data access, specialist financial agents, PII handling, audit logging, memory, and observability.

TECHNICAL SKILLS

Languages: Python, SQL
AI / LLM Systems: Agentic AI, Agentic RAG, Multi-Agent Systems, Agent Skills, Tool Calling, Structured Outputs, MCP, Context & Memory Management, Human-in-the-Loop
Backend & Data: FastAPI, Pydantic, REST APIs, PostgreSQL/SQLite, Redis, Docker, Git
Evaluation & Observability: LLM Evaluation, LLM-as-Judge, Retrieval Metrics, Citation Coverage, Token/Cost Tracking, Latency Tracking, Trace Logging, Prompt Caching